

Superstore Sales Analysis

1. Project Overview

The Superstore Sales Analysis project focuses on analysing retail sales data to identify business trends, product performance, and profitability insights. The analysis was performed using SQL, Power BI and Python to clean, transform, and visualize the data effectively.

The main objective of this project was to help understand how different categories, regions, and customer segments contribute to overall sales and profit, and to identify areas where the business can improve performance.

2. Dataset Summary

- Rows: - 9994
- Columns: - 21
- Key Records:
 - Order ID
 - Customer Name
 - Product Category
 - Sales
 - Profit
 - Quantity
 - Region
 - State
 - Shipping Mode
 - Order Date

The dataset was cleaned before analysis to remove duplicates, handle missing values, and standardize data formatting.

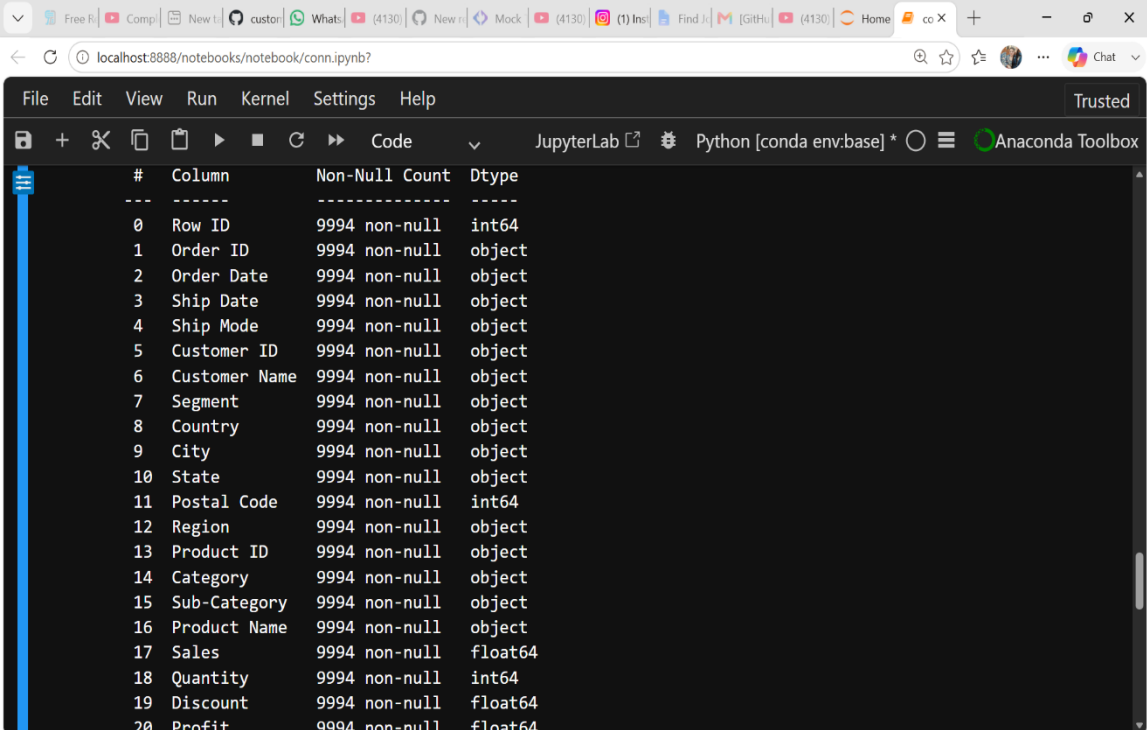
3. Business Problems Analysed

The project focused on answering the following business questions:

- 1) Which product categories generate the highest sales?
- 2) Which regions contribute the most profit?
- 3) What is the monthly sales trend?
- 4) Which customer segments are the most profitable?
- 5) How does shipping mode affect profit?
- 6) Which sub-categories perform best and worst?
- 7) Does high discount increase sales?
- 8) Which customers purchase repeatedly?

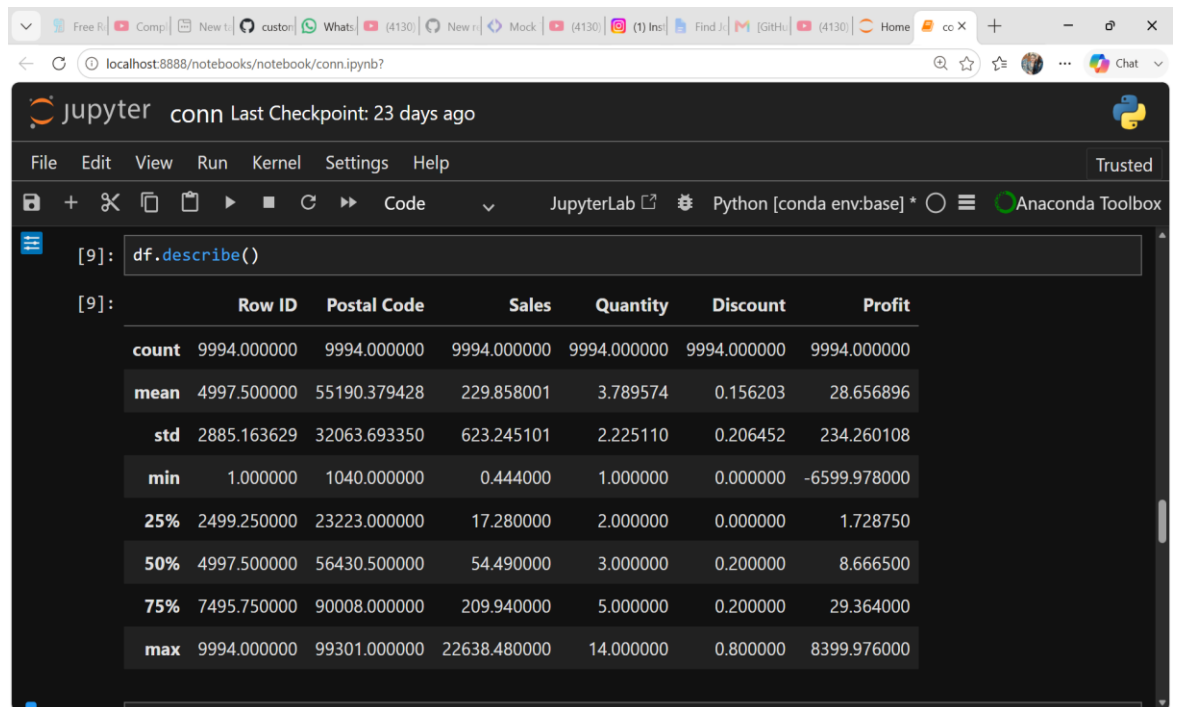
4. Cleaning Data & Missing Values

- **Data Loading:** Importing dataset using **Pandas**.
- **Initial Exploratory:** Used **df.info()** function to check structure, **df.describe()** for summarize statistics and **df.null()** function to check any missing values in dataset.



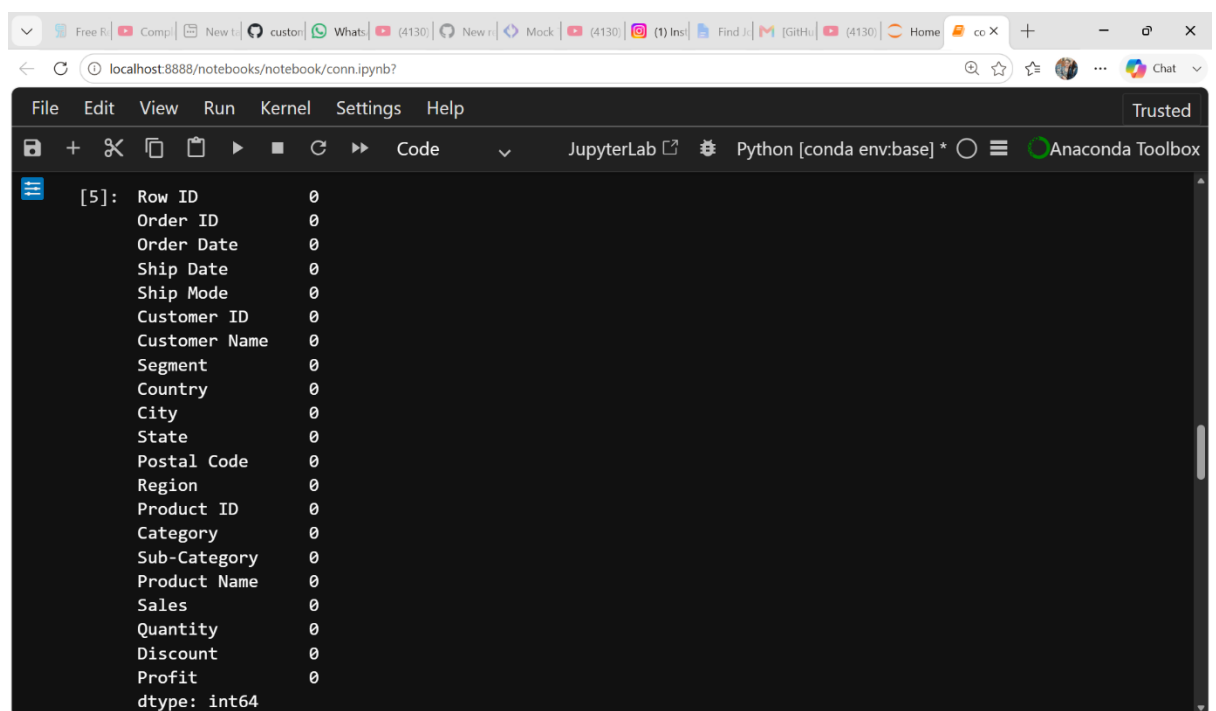
The screenshot shows a JupyterLab notebook interface with a browser window at the top displaying the URL `localhost:8888/notebooks/notebook/conn.ipynb`. The notebook itself has a dark theme and shows the output of the `df.info()` function. The output is a table with 5 columns: #, Column, Non-Null Count, Dtype, and a separator line. It lists 21 columns, all with a count of 9994 non-null values. The data types are: int64 for Row ID, Postal Code, Quantity, and Profit; object for Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Region, Product ID, Category, Sub-Category, Product Name, and Discount; and float64 for Sales.

#	Column	Non-Null Count	Dtype
0	Row ID	9994 non-null	int64
1	Order ID	9994 non-null	object
2	Order Date	9994 non-null	object
3	Ship Date	9994 non-null	object
4	Ship Mode	9994 non-null	object
5	Customer ID	9994 non-null	object
6	Customer Name	9994 non-null	object
7	Segment	9994 non-null	object
8	Country	9994 non-null	object
9	City	9994 non-null	object
10	State	9994 non-null	object
11	Postal Code	9994 non-null	int64
12	Region	9994 non-null	object
13	Product ID	9994 non-null	object
14	Category	9994 non-null	object
15	Sub-Category	9994 non-null	object
16	Product Name	9994 non-null	object
17	Sales	9994 non-null	float64
18	Quantity	9994 non-null	int64
19	Discount	9994 non-null	float64
20	Profit	9994 non-null	float64



```
[9]: df.describe()
```

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000



```
[5]:
```

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0
dtype	int64

5. Data Analysis using SQL

Performed structured analysis in PostgreSQL to answer key business questions:

- (1) **Product category by highest sales:** Checking which product has highest sales.

	Category text	total_profit double precision
1	Technology	145454.94809999999
2	Office Suppli...	122490.80080000011
3	Furniture	18451.2728

(2) **Regions by most Profit:** Found which region gained highest profit.

	Region text	profit_margin_percentage numeric
1	West	14.94
2	East	13.48
3	South	11.93
4	Central	7.92

(3) **Month-wise Sales:** Compared sales trend increase or decrease by months.

	month numeric	total_sales numeric
1	11	271693.75
2	9	248989.30
3	12	248765.33
4	3	220064.65
5	8	209964.37
6	10	184356.33
7	5	166420.32
8	7	161227.10
9	1	161083.59
10	4	147031.26
11	6	144883.50
12	2	132721.36

(4) **Customer Segment by profit:** Found which customer segment generated highest profit.

	Segment text	total_profit numeric
1	Consumer	134119.21
2	Corporate	91979.13
3	Home Offi...	60298.68

(5) Shipping mode by profit: classified shipping mode into Standard class, Second class, First class and Same day.

	Ship Mode text	total_profit numeric
1	Standard Class	164088.79
2	Second Class	57446.64
3	First Class	48969.84
4	Same Day	15891.76

(6) Sub-Categories by Performance: Identified which categories performed best or worst.

	Sub-Category text	total_sales numeric
1	Phones	330007.05
2	Chairs	328449.10
3	Storage	223843.61
4	Tables	206965.53
5	Binders	203412.73
6	Machines	189238.63
7	Accessories	167380.32
8	Copiers	149528.03
9	Bookcases	114880.00
10	Appliances	107532.16

(7) Sales increase by discount: Identified that higher discounts led to higher sales.

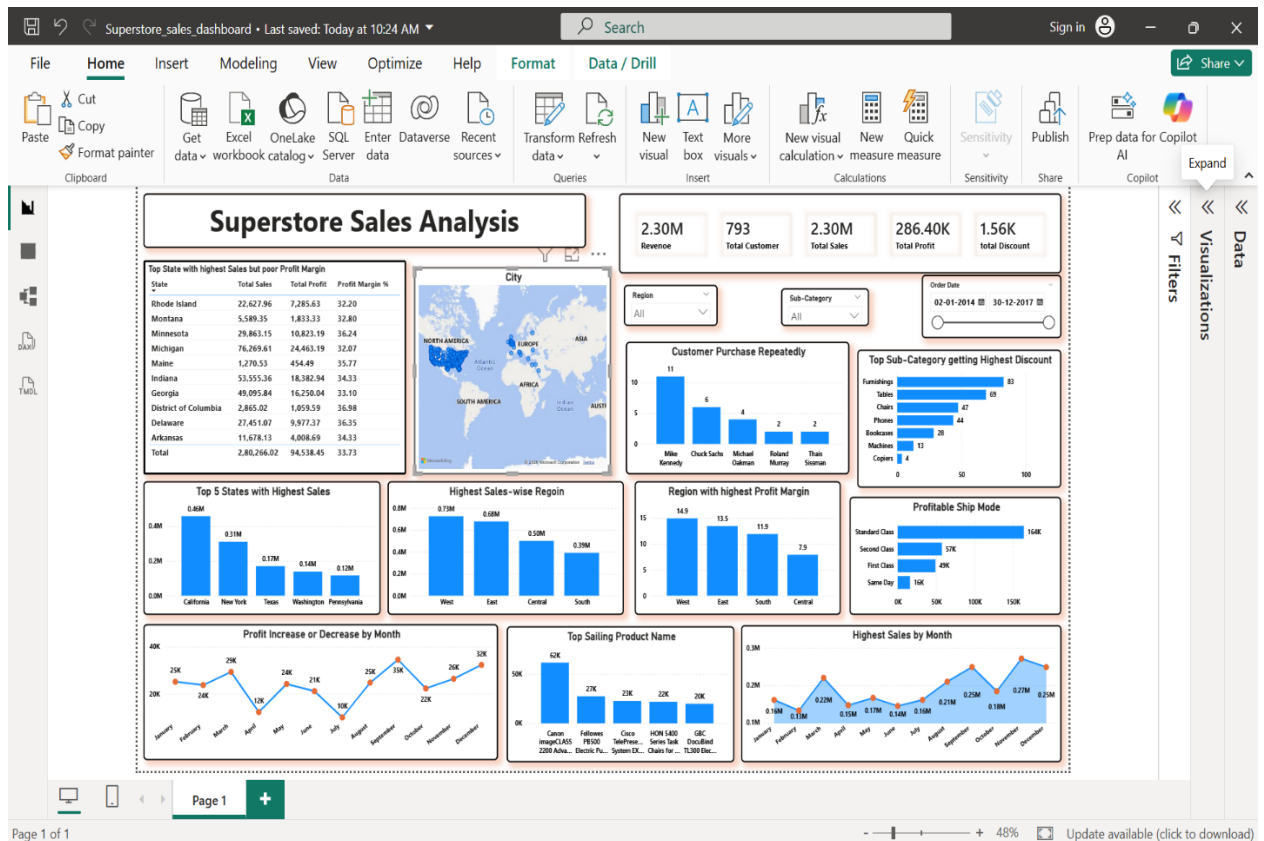
	Discount double precision 	total_sales numeric 
1	0.8	16963.76
2	0.7	40620.28
3	0.6	6644.70
4	0.5	58918.54
5	0.45	5484.97
6	0.4	116417.78
7	0.32	14493.46
8	0.3	103226.66
9	0.2	764594.37
10	0.15	27558.52
11	0.1	54369.35
12	0	1087908.47

(8) **Customer by Repeatedly Purchase:** Analysed customer purchase patterns to identify repeat customers.

	Customer ID text	Customer Name text	total_order bigint
1	WB-21850	William Brown	37
2	MA-17560	Matt Abelman	34
3	PP-18955	Paul Prost	34
4	JL-15835	John Lee	34
5	CK-12205	Chloris Kastensmidt	32
6	SV-20365	Seth Vernon	32
7	JD-15895	Jonathan Doherty	32
8	EH-13765	Edward Hooks	32
9	ZC-21910	Zuschuss Carroll	31
10	EP-13915	Emily Phan	31
11	AP-10915	Arthur Prichep	31
12	LC-16870	Lena Cacioppo	30
13	Dp-13240	Dean percer	29
14	GT-14710	Greg Tran	29
15	SH-19975	Sally Hughsby	29
16	KL-16645	Ken Lonsdale	29
17	BM-11650	Brian Moss	29
18	KD-16495	Keith Dawkins	28
19	KM-16720	Kunst Miller	28
Total rows: 788		Query complete 00:00:00.206	

6. Dashboard in Power BI

Finally, we build an interactive Dashboard in **Power BI** to represent insights visually.



7. Business Recommendation

- Reduce excessive discounts on low-profit products to improve overall profit margins.
- Improve sales strategies in low-performing regions to increase profitability.
- Optimize shipping methods to reduce operational and delivery costs.
- Monitor product-level profitability regularly to improve pricing strategies.
- Analyse customer purchasing behavior to create targeted marketing campaigns.